Better Movement Better Life

MEDICOS MINDS

Study Smart Move Better Live Healthy

THE SCIENCE OF MOVEMENT

ELECTROTHERAPY

PHYSIOTHERAPY MODALITIES FOR REHABILITATION

Reduce Pain. Restore Function. Improve Quality of Life.

Small Current... Big Changes!



WHAT IS ELECTROTHERAPY?

Electrotherapy is the therapeutic use of electrical energy in the form of low or high frequency current to treat pain, reduce inflammation, maintain or improve muscle function, and promote tissue healing.

MODALITIES = MANY USES ONE GOAL = BETTER REHABILITATION

- **TENS** (Transcutaneous Electrical Nerve Stimulation)
- **NMES** (Neuromuscular Electrical Stimulation)
- **IF** (Interferential Therapy)
- **US** (Ultrasound Therapy)
- **Iontophoresis**
- **LASER Therapy**
- **SWD** (Short Wave Diathermy)
- **MWD** (Microwave Diathermy)
- **Diathermy**

Right Modality Right Indication Better Results

KEY BENEFITS

- Pain Relief
- Reduced Inflammation & Edema
- Improved Circulation
- Muscle Re-education
- Tissue Healing
- Maintains Muscle Mass

Science + Skill = Recovery

FOR MEDICAL STUDENTS

- MBBS
- BHMS
- BSc. NURSING
- BDS
- BPT

Concepts for Exams & Clinical Practice



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LEARN | PRACTICE | HEAL

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HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Electrotherapy

Author: Dr. MD Kamran Alam | Instagram: @restore_healthphysio

OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.

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4. LOW FREQUENCY CURRENTS

DEFINITION :

Low frequency currents (LFC) are Electrical currents with a frequency range of 1 Hz to 1000 Hz (1 kHz). They produce neuromuscular stimulation, pain relief and other therapeutic effects depending on the type of current used.

TYPES OF LOW FREQUENCY CURRENTS

1. Faradic Current (0.1 – 10 Hz)
2. Galvanic Current (Direct Current)
3. Interferential Current (IFC)
4. Premodulated Current (Pre-Mod)
5. TENS (Transcutaneous Electrical Nerve Stimulation)
6. Russian Current
7. Diadynamic Currents (MF, DF, CP, LP)

GENERAL PRINCIPLE :

Low frequency currents act by stimulating nerves and muscles or by producing chemical and thermal changes in tissues depending on the type, duration and intensity of current.

DURATION :

Generally 10 – 20 minutes (according to condition and type of current used).

THERAPEUTIC EFFECTS :

- Pain relief (analgesic effect)
- Muscle stimulation and strengthening
- Prevention or retardation of disuse atrophy
- Increase in local blood circulation
- Reduction of muscle spasm
- Maintenance or increase of range of motion
- Facilitation of healing
- Improve muscle re-education and coordination

FREQUENCY RANGE

Type	Frequency (Hz)
Faradic	0.1 – 10
Interferential	1 – 200
TENS	1 – 150
Russian	2500
Diadynamic	50 – 100

INDICATIONS :

- Musculoskeletal pain
- Muscle weakness and atrophy
- Post-operative muscle re-education
- Neuromuscular re-education
- Edema (depending on current)
- Peripheral nerve injuries
- Chronic pain conditions
- Sport injuries

CONTRAINDICATIONS :

- Cardiac pacemaker
- Malignancy
- Epilepsy
- Pregnancy (over abdomen and low back)
- Active bleeding or hemorrhage
- Thrombophlebitis
- Skin infection or open wounds
- Impaired sensation

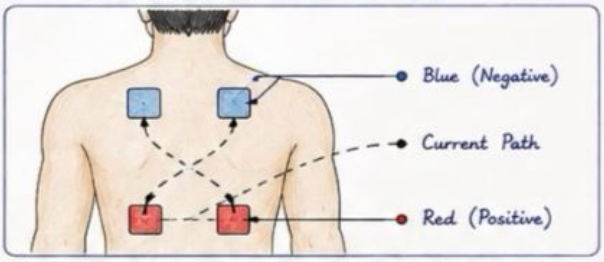
COMMON LOW FREQUENCY CURRENTS – KEY POINTS

1. **FARADIC CURRENT** : Used for muscle stimulation in complete denervation. Frequency 0.1 – 10 Hz (burst type).
2. **GALVANIC CURRENT** : DC current used for iontophoresis, pain relief (using polarity) and edema reduction.
3. **INTERFERENTIAL CURRENT** : Two medium frequency currents intersect to produce a low frequency effect. Good for pain relief and muscle stimulation.
4. **PREMODULATED CURRENT** : Medium frequency current modulated with low frequency. Used for deep pain relief and muscle stimulation.
5. **TENS** : Used mainly for pain relief via gate control theory. Frequency 1 – 150 Hz.
6. **RUSSIAN CURRENT** : High frequency (2500 Hz) used for muscle strengthening and re-education.
7. **DIADYNAMIC CURRENTS** : MF, DF, CP, LP – used for pain relief, muscle stimulation and edema management.

PROCEDURE (General Steps) :

1. Explain the procedure and take consent.
2. Check the patient's condition and screen for contraindications.
3. Inspect the skin and clean the area if needed.
4. Select the appropriate current, frequency, intensity and duration.
5. Apply electrodes with proper size, type and placement. (Use electrode cream or wet sponges if required.)
6. Increase intensity gradually to the therapeutic level.
7. Monitor the patient's response throughout the treatment.
8. Reduce intensity slowly and turn off the machine.
9. Remove electrodes, clean the skin and assess response.
10. Document the treatment.

ELECTRODE PLACEMENT EXAMPLE



(Example for pain relief using IFC / Pre-mod)

PRECAUTIONS :

- Use correct parameters for each condition.
- Always increase intensity gradually.
- Do not use over impaired sensation areas.
- Avoid placing electrodes over carotid sinus, eyes, or heart area.
- Keep the skin clean and dry.

ADVANTAGES :

- Easy to use and relatively safe.
- Wide range of therapeutic benefits.
- Effective for both pain relief and muscle stimulation.
- Can be used in acute and chronic conditions.
- Helps in faster recovery and functional improvement.

THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

SUPPORT & COMMUNITY:

- Instagram: @restore_healthphysio (Daily Clinical Case Studies & Quizzes)
- WhatsApp Helpline: +91 83407 49923
- Email Support: restorehealthphysio@gmail.com

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